

# MANITHA PASANDUL

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 [Manitha Pasandul](#) |  [Portfolio](#)

Bandarawatta, Gampaha - 11000, Sri Lanka

## OBJECTIVE

Final-year Aeronautical Engineering undergraduate at the University of Moratuwa with experience in aerodynamic design, flight dynamics, CFD, structural analysis, and aircraft maintenance. Skilled in ANSYS Fluent, MATLAB, SolidWorks, and XFLR5, with practical exposure through research projects, Formula Student aerodynamics, and industrial training. As a prospective PhD candidate, I am interested in aerospace propulsion, space science and engineering, hypersonic aerodynamics, aeroelasticity, flight dynamics, and computational aerodynamics.

## EDUCATION

- **University of Moratuwa** 2022 - 2026  
*BSc. (Hons) in Mechanical Engineering - Aeronautical Stream* Katubedda, Sri Lanka
  - CGPA: 3.80/4.00
- **Bandaranayake College** 2012 - 2020  
*GCE Advanced Level Examination, Physical Science Stream* Gampaha, Sri Lanka
  - Z-score: 2.1187
  - GCE Ordinary Level Examination (2017): 9 As
- **Sri Bodhi Primary College** 2007 - 2011  
*Primary Education* Gampaha, Sri Lanka

## WORK EXPERIENCE

- **Temporary Instructor** July 2026 – Present  
*Department of Mechanical Engineering, University of Moratuwa, Moratuwa, Sri Lanka*
  - Support the delivery of tutorials, practical sessions, and laboratory activities for Manufacturing Processes, Basic Thermodynamics, Mechanics of Machines I, and Principles of Naval Architecture
  - Assist academic staff with laboratory preparation, practical demonstrations, student supervision, and the coordination of course-related activities
- **Trainee Aeronautical Engineer** Jan 2025 – Jun 2025  
*SriLankan Airlines, Bandaranaike International Airport, Katunayake, Sri Lanka*
  - Gained hands-on exposure through workshops on engine modules, wheels and brakes, NDT, paint and composite repairs, and structural maintenance
  - Developed practical understanding of aircraft sub-systems by observing and assisting inspection, disassembly, repair, and reassembly processes
- **Aerodynamics Engineer** Dec 2025 – Present  
*FalconE Racing Team*
  - Designed and optimized aerodynamic components using CFD analysis to improve downforce, drag performance, and manufacturability
  - Supported structural integration and carbon fiber composite fabrication of aerodynamic components for the Formula Student race car

## PROJECTS

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### • Framework for Flight Dynamics and Control of Small Surveillance UAVs

Aug 2025 – June 2026

Tools: AVL, MATLAB, SolidWorks, Flow5

- Developed a framework to analyze flight dynamics and control characteristics of small conventional surveillance UAVs
- Performed sensitivity analysis of geometric parameters to study their effect on aircraft dynamic modes and stability behaviour
- Fabricated a 2.5 m span UAV and conducted flight testing to experimentally validate modal characteristics
- Compared simulated and experimental modal results to identify deviations and improve the reliability of the analysis framework

### • Aircraft Propeller Design and Performance Analysis

Jun 2025 – Dec 2025

Tools: MATLAB, SolidWorks

- Designed a two-bladed aircraft propeller using blade element theory and iterative MATLAB calculations
- Determined blade geometry and estimated thrust, torque, inflow factors, efficiency, and required engine power

### • CFD Analysis of NREL S809 Aerofoil

June 2025 – Dec 2025

Tools: ANSYS Fluent, MATLAB

- Performed 2D steady-state CFD analysis of the NREL S809 aerofoil for wind turbine blade applications
- Evaluated lift, drag, lift-to-drag ratio, pressure contours, and velocity contours for angles of attack from 0° to 20°

### • Reverse Engineering of Aircraft Parts Using FAROARM and CAM

Jan 2026 – June 2026

Tools: FAROARM, Geomagic Design X, SolidWorks, SolidWorks CAM

- Reverse engineered an aircraft component using FAROARM point cloud data and reconstructed the CAD model in SolidWorks
- Generated CAM toolpaths, machining simulation, and CNC G-code while evaluating dimensional deviations and manufacturing limitations

### • Experimental and Numerical Stress Analysis of a Wing Structure

Jan 2026 – June 2026

Tools: ANSYS Static Structural

- Developed an ANSYS finite element model to predict stress and strain distributions under static loading
- Validated FEA results using experimental strain gauge measurements and identified possible sources of deviation

### • Design of a Sailplane Wing

Aug 2024 – Dec 2024

Tools: XFLR5, MATLAB, SolidWorks

- Designed a high-performance sailplane wing through aerofoil selection, finite-wing analysis, and glider performance calculations
- Designed a dual-spar tapered wing structure and evaluated stress, deflection, shear flow, buckling, and material selection

### • Design and Development of Front Wing Aerodynamic Package

Jan 2025 – Dec 2025

Tools: Ansys Fluent, SolidWorks

- Designed a four-element front wing and optimized its aerodynamic geometry to improve downforce while maintaining structural feasibility
- Developed structural components and fabricated the final wing using carbon fiber composite layup techniques for lightweight and high-stiffness performance

HONORS AND AWARDS

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|-----------------------------------------------------------------------------------------|------------------|
| • <b>Dean’s List - Semester 2</b><br><i>University of Moratuwa   Semester GPA: 3.86</i> | <i>July 2023</i> |
| • <b>Dean’s List - Semester 4</b><br><i>University of Moratuwa   Semester GPA: 3.91</i> | <i>Aug 2024</i>  |
| • <b>Dean’s List - Semester 6</b><br><i>University of Moratuwa   Semester GPA: 4.00</i> | <i>Aug 2025</i>  |
| • <b>Dean’s List - Semester 7</b><br><i>University of Moratuwa   Semester GPA: 3.85</i> | <i>Dec 2025</i>  |
| • <b>Dean’s List - Semester 8</b><br><i>University of Moratuwa   Semester GPA: 4.00</i> | <i>July 2026</i> |

VOLUNTEER EXPERIENCE

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|-----------------------------------------------------------------------------|----------------------------|
| • <b>Volunteer Teacher (Ganitha Sawiya)</b><br><i>Sasnaka Sansadaya</i>     | <i>2022 – 2023</i>         |
| • <b>Volunteer - Aerodynamics Team Member</b><br><i>FalconE Racing Team</i> | <i>Aug 2024 – Dec 2025</i> |
| • <b>Demonstrator at Techno 2024</b><br><i>FalconE Racing Team</i>          | <i>Oct 2024</i>            |

PROFESSIONAL MEMBERSHIPS

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|-----------------------------------------------------------------------------------------------------------|---------------------------|
| • <b>Student Member</b> Institution of Engineers Sri Lanka (IESL)<br>Membership No: S-30208               | <i>Mar 2024 - Present</i> |
| • <b>Student Affiliate Member</b> Institution of Mechanical Engineers (IMechE)<br>Membership No: 80652300 | <i>Dec 2022 - Present</i> |

ADDITIONAL INFORMATION

**Languages:** Sinhala (Proficient), English (Proficient)  
**Interests:** High Speed Aerodynamics, Aerospace Propulsion, Flight Dynamics and Control, Aeroelasticity

REFERENCES

1. **Dr. Nalaka Samaraweera**  
Senior Lecturer, Department of Mechanical Engineering  
University of Moratuwa  
Email: nalakas@uom.lk  
Relation: Project Supervisor